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Final Report:
Telco Customer Churn Predictive Modeling

Problem Statement:

Customer churn presents a significant challenge for telecommunications companies because losing existing customers directly impacts recurring revenue and long-term profitability. While telecom providers collect large amounts of customer data, it is often difficult to determine which factors most strongly influence churn and which customers are most likely to leave.

Understanding these patterns is essential for businesses that want to proactively retain customers rather than reacting after cancellations occur.

Background:

Customer acquisition costs in the telecommunications industry are high, making customer retention a critical component of sustainable growth. Research consistently shows that retaining existing customers is far more cost-effective than acquiring new ones. As a result, telecom companies increasingly rely on data analytics to better understand customer behavior and reduce churn.

Customer churn is influenced by a variety of factors including pricing, service contracts, tenure with the company, and additional service offerings. However, identifying the most important drivers of churn requires systematic data analysis and predictive modeling. By analyzing customer characteristics and service usage patterns, companies can identify at-risk customers and implement targeted retention strategies before those customers cancel their services.

Approach:

The goal of this project is to analyze customer data from a telecommunications provider to identify the key drivers of churn and build predictive models capable of identifying customers who are likely to leave. The project follows a structured data science workflow consisting of data cleaning, exploratory data analysis, preprocessing, and predictive modeling.

First, the dataset was cleaned and prepared to ensure that all variables were correctly formatted and free from missing or inconsistent values. Next, exploratory data analysis (EDA) was conducted to examine distributions of individual features and relationships between customer attributes and churn. Then preprocessing was done in order to encode variables and split the data for the models. Finally, several machine learning models were trained and evaluated to determine which approach best predicts customer churn.

The overall objective is not only to build an accurate predictive model but also to generate actionable insights that telecommunications companies can use to improve customer retention strategies.

Data Wrangling:

Before performing analysis or modeling, the dataset required several data wrangling steps to ensure data quality and consistency. The first step involved inspecting the dataset for missing values, incorrect data types, and inconsistencies within the variables.

One important issue identified during this process involved the **TotalCharges** variable. Although this column represents numerical billing information, it was initially stored as a string. This occurred because some records contained blank entries for customers who had very short tenure. These values were converted to numeric format, and rows containing invalid or missing entries were removed to ensure accurate analysis.

After correcting data types, additional checks were conducted to verify that all features were appropriately formatted. Categorical variables such as

Contract, **InternetService**, and **PaymentMethod** were separated from numerical variables like **tenure**, **MonthlyCharges**, and **TotalCharges**. This separation allowed the dataset to be properly prepared for the machine learning pipeline, where categorical variables would later be encoded and numerical variables scaled.

Finally, the dataset was reviewed for duplicate records and overall data integrity. Once these data wrangling steps were completed, the dataset was considered clean and ready for exploratory analysis and modeling.

Exploratory Data Analysis:

Exploratory data analysis was conducted to understand the structure of the dataset and identify patterns related to customer churn. The analysis focused on both single-feature distributions and feature interaction relationships with churn.

Single Feature Analysis

Several visualizations were created to explore the distributions of individual variables within the dataset. Significant ones include:

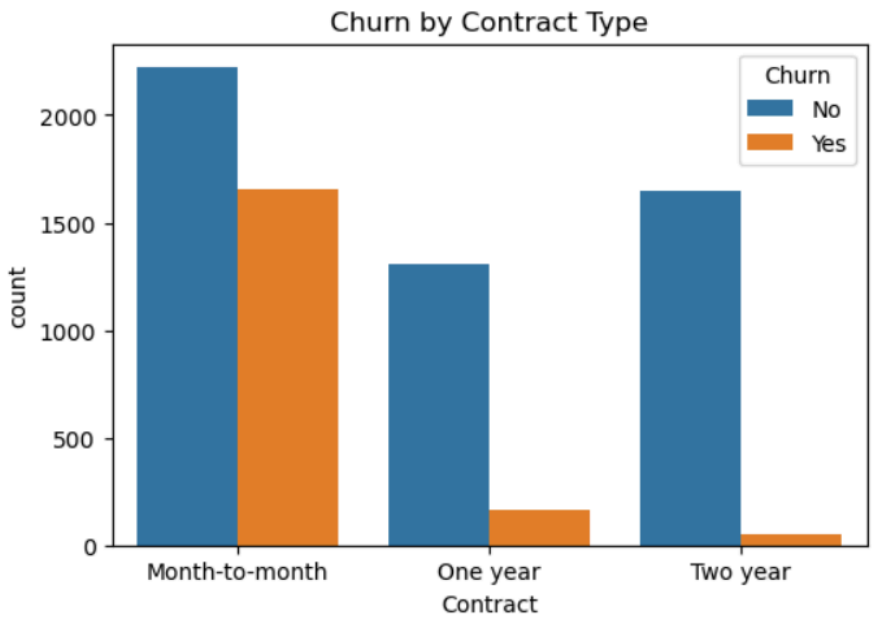

Figure 1.

Figure 1: Placing churn by contract type on a bar graph shows a significantly higher churn rate for those who are on a monthly contract compared to those on a yearly contract.

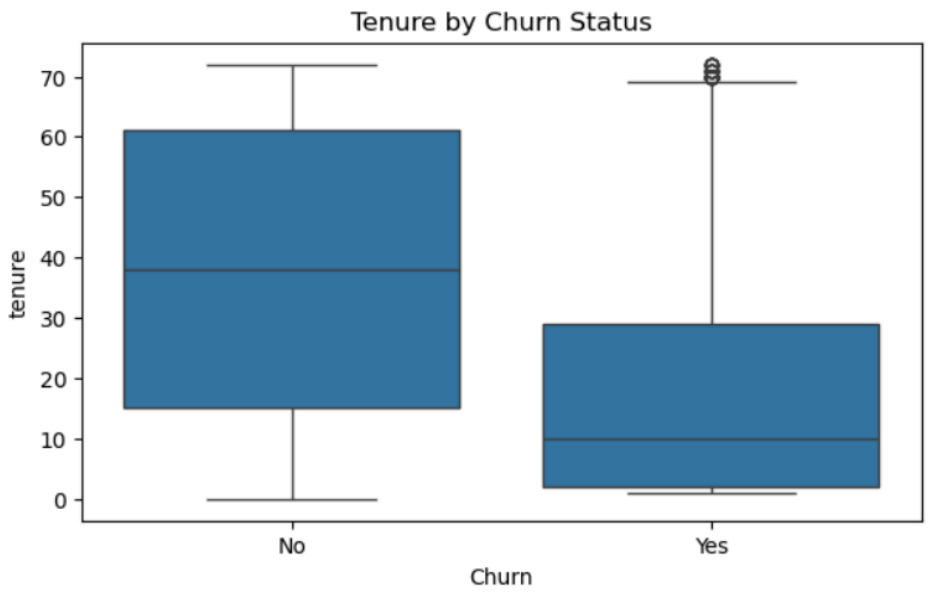

Figure 2.

Figure 2: Tenure, represented in the number of months a customer has been with the company, is shown to have a direct correlation with churn status. As tenure grows, the rate of churn decreases.

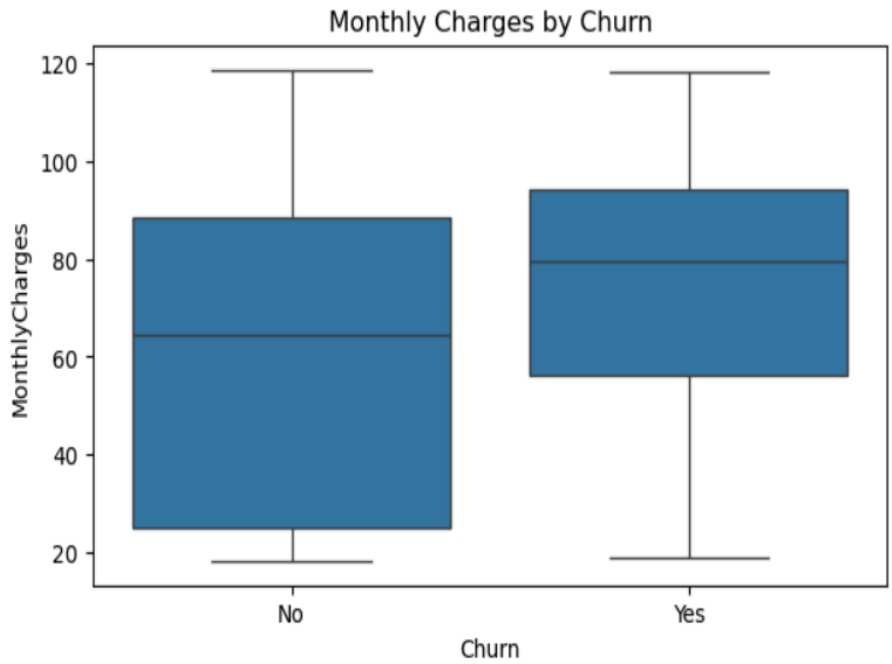

Figure 3.

Figure 3: This boxplot shows that a higher monthly charge leads to increased risk of churn.

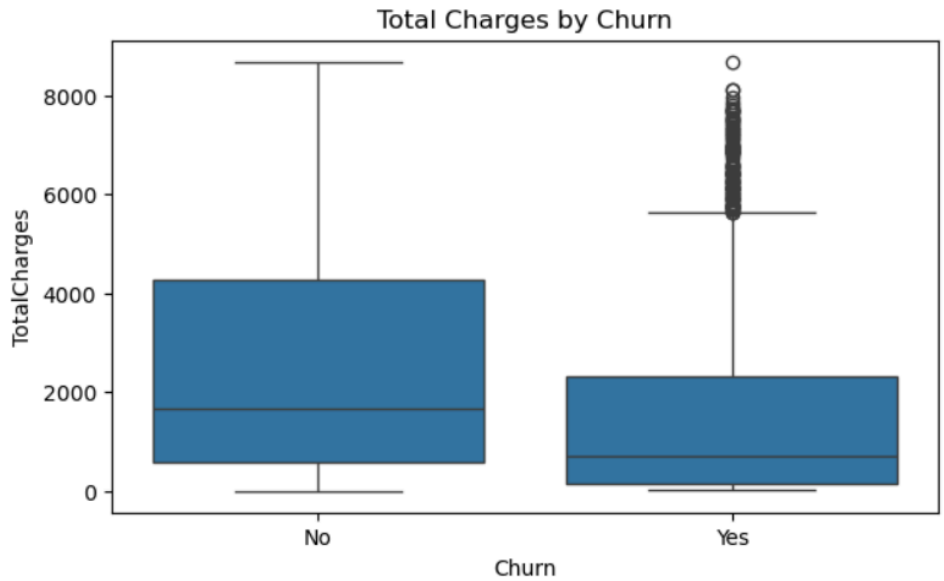

Figure 4.

Figure 4: This shows customers who churn tend to have significantly lower lifetime spending, suggesting that attrition occurs more frequently among newer customers rather than long-term subscribers.

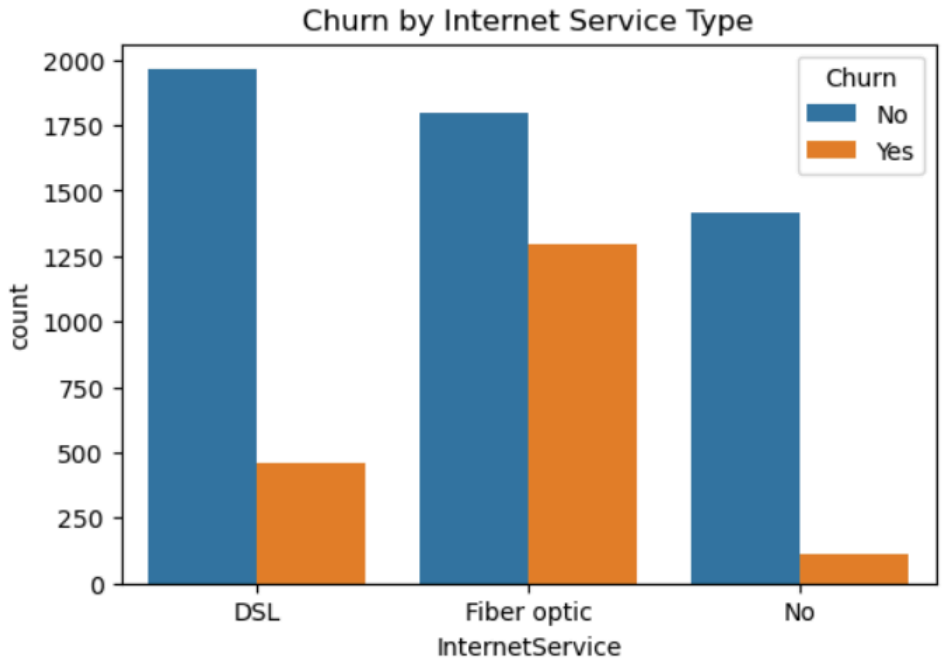

Figure 5.

Figure 5: This shows fiber optic customers exhibit the highest churn rates, suggesting that higher-paying, high-speed users may also have higher expectations and greater sensitivity to pricing or service quality.

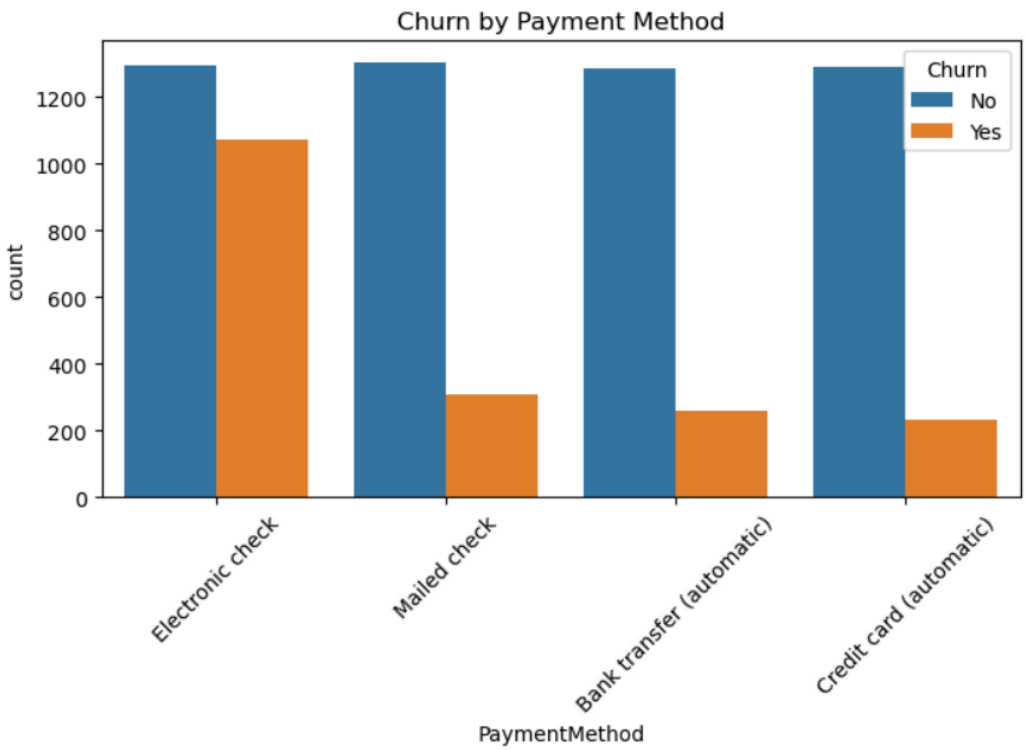

Figure 6.

Figure 6: This shows customers paying with electronic checks are much more likely to leave, while those on automatic payments tend to stay. This suggests that customers who are more 'locked in' through autopay are more stable

Feature Interaction Analysis

While the previous analysis examined each feature independently, customer churn behavior may be influenced by combinations of features rather than single variables alone. To explore this, we analyze interaction effects by grouping customers across pairs of related features and observing differences in churn rates. Noteworthy ones include:

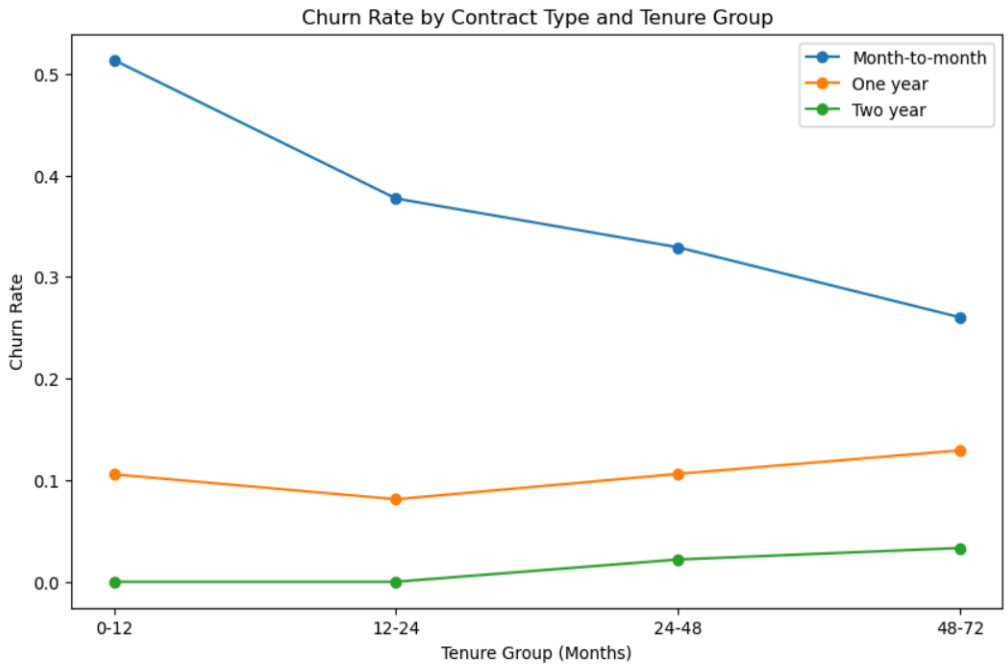

Figure 7.

Figure 7: Customers on month-to-month contracts exhibit the highest churn rates, particularly in early tenure periods. However, even long-tenure customers on month-to-month plans churn at higher rates than customers on longer-term contracts. This suggests that contract structure has a persistent impact on churn risk beyond just customer age.

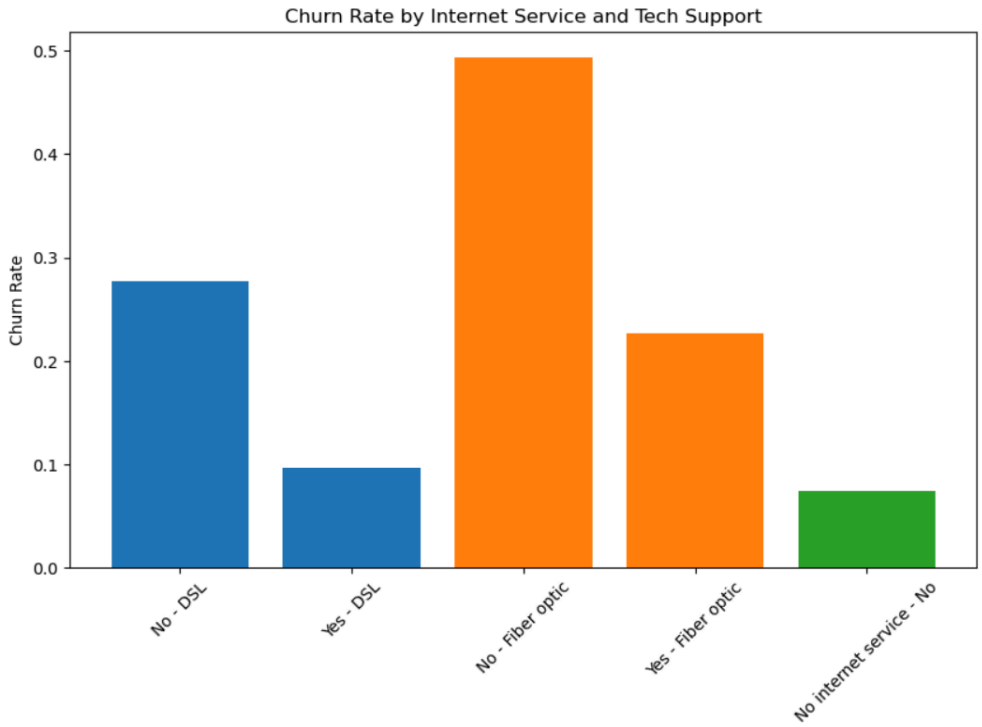

Figure 8.

Figure 8: Customers with fiber optic service who do not have tech support demonstrate significantly higher churn rates compared to those who do. This suggests that higher-tier internet customers may require additional service stability and support, and lack of technical assistance may contribute to dissatisfaction and churn.

Key EDA Insights

The exploratory analysis revealed several important patterns:

- Customers on month-to-month contracts are significantly more likely to churn
- Shorter tenure customers show higher churn rates
- Higher monthly charges may be associated with increased churn risk

These insights guided the modeling phase by highlighting which variables were most likely to contribute to predictive performance.

Modeling:

To predict customer churn, three classification models were trained and evaluated: Logistic Regression, Random Forest, and Gradient Boosting. Each model was trained using a standardized preprocessing pipeline that scaled numerical features and encoded categorical variables.

Model Performance

Model	Accuracy	ROC-AUC
Logistic Regression	0.799	0.842
Random Forest	0.789	0.823
Gradient Boosting	0.796	0.842

Logistic Regression achieved the highest performance across both evaluation metrics. Gradient Boosting performed nearly identically in terms of ROC-AUC but slightly lower in accuracy, while Random Forest produced the weakest results among the three models.

ROC-AUC was used as a key evaluation metric because it measures the model's ability to distinguish between churners and non-churners across all classification thresholds. A score above **0.84** indicates strong predictive capability.

Model Comparison

Although ensemble models such as Random Forest and Gradient Boosting are often capable of capturing complex nonlinear relationships, the results indicate that Logistic Regression performed slightly better despite being the simplest model.

This suggests that churn in this dataset is primarily influenced by additive, structured relationships between customer features, rather than complex nonlinear interactions. As a result, the simpler model was able to capture the predictive signal effectively without requiring additional model complexity.

Another advantage of Logistic Regression is its interpretability. The model's coefficients allow analysts to directly understand how different variables influence churn probability, making it easier for stakeholders to translate model outputs into business strategies.

For these reasons, Logistic Regression was selected as the final model.

Findings/Takeaways:

The results of this analysis demonstrate that customer churn can be predicted with strong reliability using customer service data and behavioral attributes. With a ROC-AUC score of approximately **0.84**, the predictive model can effectively distinguish customers who are at risk of leaving from those who are likely to remain.

Several key drivers of churn were identified during the analysis. Customers on **month-to-month contracts**, customers with **short tenure**, and customers with **higher monthly charges** appear to be more likely to churn. These findings suggest that early customer engagement and contract structure play a significant role in retention.

Since Logistic Regression performed best, churn appears to be driven by consistent, additive factors (e.g., contract type, tenure, pricing variables), rather than unpredictable or highly complex patterns. That makes the problem easier to manage operationally.

For telecommunications companies, these insights can be translated into actionable strategies. Businesses can implement churn prediction systems to identify high-risk customers and proactively offer retention incentives,

service improvements, or contract upgrades. Encouraging customers to transition from flexible month-to-month contracts to longer-term agreements may also reduce churn rates.

Ultimately, this analysis demonstrates how data-driven decision making can help telecommunications providers improve customer retention and reduce revenue loss.

Further Research:

Although this project produced valuable insights, several opportunities exist to expand the analysis in future research.

One potential improvement would involve incorporating additional customer behavior features such as service usage patterns, customer support interactions, or billing history. These variables could provide deeper insight into the behavioral signals that precede churn.

Another potential extension would involve testing more advanced machine learning models, despite our conclusion that a simple linear model was best. Optimized implementations such as XGBoost or LightGBM, may capture more complex feature interactions and potentially improve predictive performance.

Future research could also explore time-based modeling approaches, which analyze churn risk over time rather than treating the problem as a single snapshot. Survival analysis or time-series modeling techniques could help predict not only whether a customer will churn, but when churn is most likely to occur.

Finally, additional segmentation analysis could be conducted to examine churn patterns across different customer groups, allowing companies to develop more targeted retention strategies.